

Zero Hunger - AI Crop Disease Detection Device



HARI ADITHYA V(LEAD) (II Year - CSE)

SABARISH R S (III Year - ECE)

BAALAMURUGA Y K(II Year - CSE)

ARUNKUMAR P(II Year - CSE)

TEAM NAME:The Ideators

1.Dr.M. Jothibasu

Assistant professor,
ECE

2.Dr.R.Ravikumar

Assistant professor,
Science and Humanities

2.Zero Hunger - AI Crop Disease Detection Device



In today's agricultural landscape, **farmers face severe crop losses due to late detection of plant diseases.** By the time visible symptoms appear, the infection has already damaged a large portion of the crop, leading to **reduced yield, poor quality produce, and financial instability for farmers.**

Risks faced by farmers:

- **Late diagnosis leads to rapid disease spread, resulting in yield losses of up to 40–50% in many crops.**
- Governments and agencies lack real-time disease data for region-wide monitoring.
- Even after detecting disease, farmers struggle to find where to buy the correct medicine or who can help
- In the absence of accurate diagnosis, farmers apply chemical pesticides unnecessarily, increasing costs and harming soil health.
- Continuous use of chemicals and replanting due to disease loss raise input costs, reducing profit margins.
- Over-application of pesticides contaminates soil and water, affecting biodiversity and the surrounding ecosystem.
- **Repeated crop losses push farmers into debt and financial stress, weakening rural economies and sustainability.**

Objective:

- **To develop a portable AI-powered hardware device for early and accurate crop disease detection.**
- To use image, spectral, and environmental sensors for identifying plant stress even before visible symptoms appear.
- **To offer treatment recommendations via a connected Farmer Dashboard and AI chatbot.**
- To integrate soil and weather data for personalized and precise disease prediction.
- To perform TRL-7 field validation in collaboration with agronomists and research institutes.
- **To reduce yield loss, minimize pesticide misuse, and promote sustainable, data-driven farming.**
- To contribute toward Zero Hunger (SDG 2)

3. Idea Description:



Core Concept: A portable AI-based hardware device designed to detect crop diseases at an early stage using multimodal sensing (image, spectral, and environmental data).

Description of Idea:

- The device first captures a clear image of the crop leaf using an RGB camera, along with sensor readings such as temperature, humidity, VOC levels, and soil moisture.
- The system extracts key features from the image (color, texture, and lesion patterns) and sensor data (environmental and chemical indicators of stress).
- The processed data is passed to a trained AI model that classifies the input as Healthy, Early-Stage Disease, or Severe Infection based on learned patterns.
- The output is displayed on a mobile app, showing the disease name, confidence level, and risk score.
- Farmers can take the recommended actions and provide feedback on outcomes, which helps improve the AI model's accuracy over time.

Output & Alerts:

- Provides instant results on an mobile app with voice or text alerts in local languages.
- Daily task planner – irrigation schedule, spraying reminders, fertilizer timing.
- Suggests preventive measures or treatments based on detected disease type and severity.
- Data Integration:
 - 1.Stores and transmits data to a cloud dashboard for monitoring disease trends and regional analysis.
 - 2.Enables agronomists and researchers to validate and improve detection accuracy.



3. Innovation



Our Key Innovations

1. Multisensor + AI Hardware Fusion

A portable handheld device combining camera, spectral, gas, and soil sensors powered by Edge AI (TinyML) enables early, offline crop disease detection in real farm environments—affordable, scalable, and accessible to every farmer.

2. Offline-Capable AI Hardware

Rugged, solar-powered handheld devices and kiosks perform disease detection directly on the edge — no internet required. Farmers in remote areas receive instant results and guidance even without connectivity.

3. Smart Farmer Ecosystem

An intelligent Farmer Dashboard provides disease details, severity level, daily action steps, voice-based guidance, and organic/chemical remedy recommendations—all in the farmer's local language.

4. Green Technology Integration

Solar charging support, biodegradable casings, and modular replaceable parts extend lifespan while minimizing electronic waste.

5. Explainable & Transparent AI Decisions:

Each detection includes confidence scores and an explanation (e.g., “Early blight probability: 92%, based on leaf spot pattern + low chlorophyll index”). This builds trust and aids farmer learning.

6. Integrated AgriPharma & Supply Network

In-app access to verified nearby agro-pharmacies, available medicines, dosage instructions, and online ordering ensures timely access to safe and approved treatments.

7. Transport & Seller Connectivity

Dedicated Seller and Transport Dashboards connect farmers, suppliers, and delivery agents—managing medicine availability, logistics tracking, and route optimization for last-mile delivery.

Future Implementations

1. Community-Driven Knowledge Hub

- Farmers access a voice-enabled forum to share insights, receive expert help, and explore government schemes, insurance options, and success stories, empowering collaborative learning

2. Farmer Community and AI Education

- Build a digital farmer network for peer learning, discussions, and live Q&A with agronomists.
- Launch AI literacy workshops for rural communities to promote digital empowerment.

4. Technicalities and Scope



Technicalities

- **Core Board:** Raspberry Pi 5 / ESP32 (Edge AI microcontroller)
- **Camera Sensor:** (IMX708)
- **Spectral sensor:** AS7265x or AS7341.
- **VOC sensor:** MQ-135 or SGP30.
- **Environmental sensors:** DHT22 or BME280, capacitive soil moisture.
- **Connectivity for alerts:** SIM7600 4G modem (SMS / HTTP) OR Wi-Fi/4G U dongle OR LoRa gateway depending on area.
- **Pan mechanism:** Continuous-rotation servo or stepper motor + gearbox + turn

Software Stacks

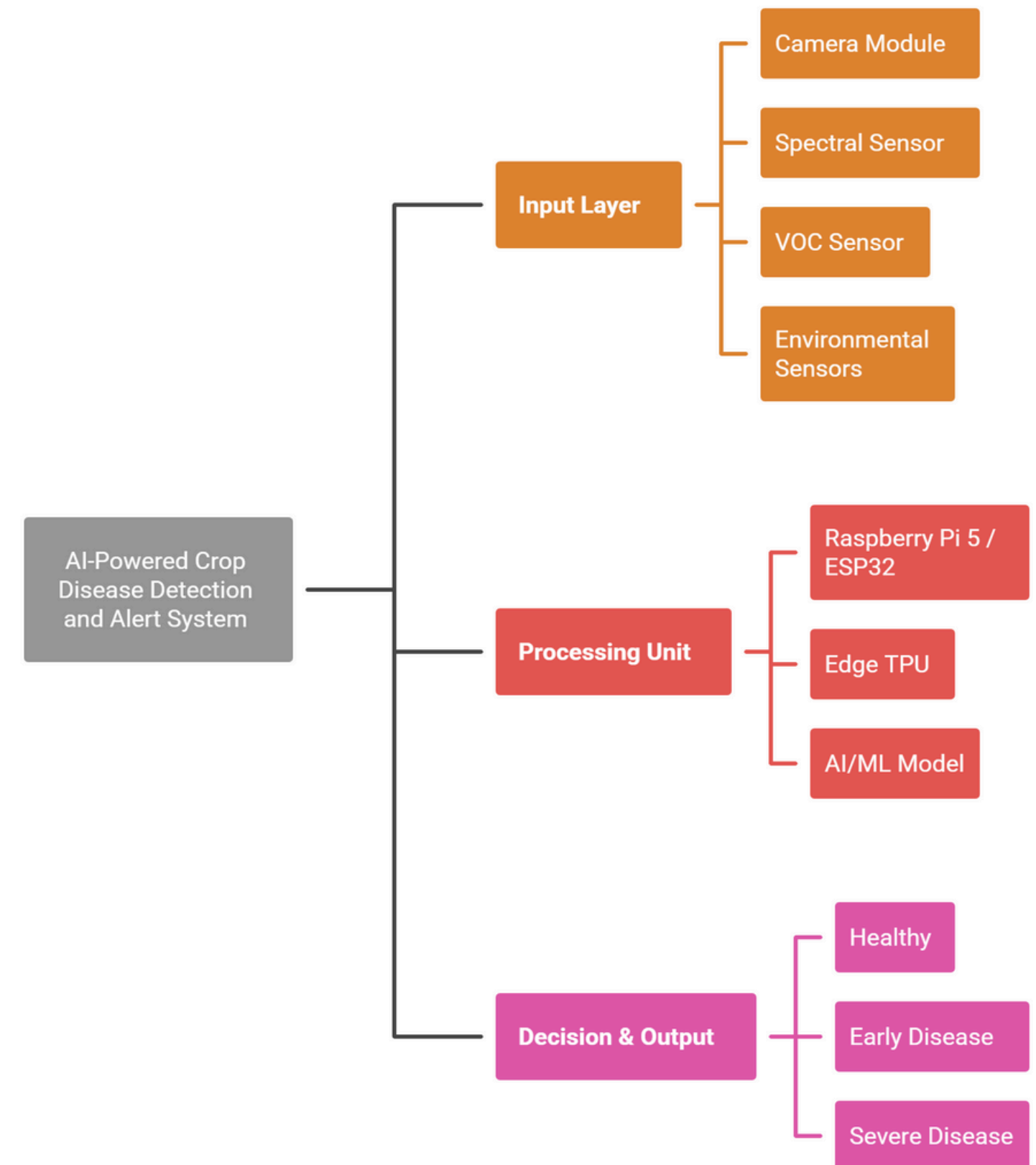
- **Farmer Dashboard (Mobile/Web):** Displays disease type, severity, cure, daily tasks, and nearby remedies.
- **AI Chatbot (Multilingual):** Voice-enabled, local-language assistant for step-by-step guidance.
- **Seller & Transport Dashboard:** Manages medicine availability, delivery tracking, and logistics.
- **Cloud Analytics Dashboard:** Heatmaps, regional trends, and predictive analytics for agronomists and officials.



TensorFlow



PyTorch



5. Target Audience/Market



Small & Marginal Farmers (Main Users)

- Problem: Lack of early detection, access to agronomists, and awareness of correct treatments.
- Affordable, easy-to-use AI device that detects disease and gives instant voice alerts in local language.

Agri-Input Dealers & Pesticide Sellers

- Problem: Demand forecasting and stock management.
- The seller dashboard predicts local demand spikes (e.g., fungal disease → fungicide sales).

Agronomists, Researchers & Agriculture Departments

- Problem: Manual disease survey is slow and inconsistent.
- Use AgriAI data for real-time disease tracking, regional reports, and training models.

AgriTech Startups / NGOs / Rural Entrepreneurs

- Problem: Need affordable, modular tech for rural deployment.
- AgriAI Guardian is open, low-cost, and scalable with startup-friendly APIs.

Farmer Producer Organizations (FPOs) & Cooperatives

- Problem: Need to monitor hundreds of fields collectively.
- Why Us: Deploy multiple devices and get regional disease heatmaps + data analytics on a single dashboard.

6. Social/Environmental Impact

Empowering Farmers with AI

- Brings cutting-edge AI tools directly to small and marginal farmers, many of whom have limited digital access.
- Enables early disease detection and reducing dependency on external agronomists.
- **Impact:** Increases farmer confidence, improves productivity, and promotes tech inclusion.

Reducing Farmer Loss & Debt

- Detecting diseases early reduces 30–40% yield loss, saving farmers from economic distress.
- **Impact:** Improves financial security and mental well-being of farming communities.

Reduction in Pesticide Pollution

- Optimized and targeted treatment recommendations reduce chemical overuse by up to 40%.
- Promotes eco-friendly alternatives (Neem oil, Trichoderma, bio-fungicides).
- **Impact:** Cleaner soil, safer food, and reduced toxic runoff into rivers.

Sustainable Resource Management

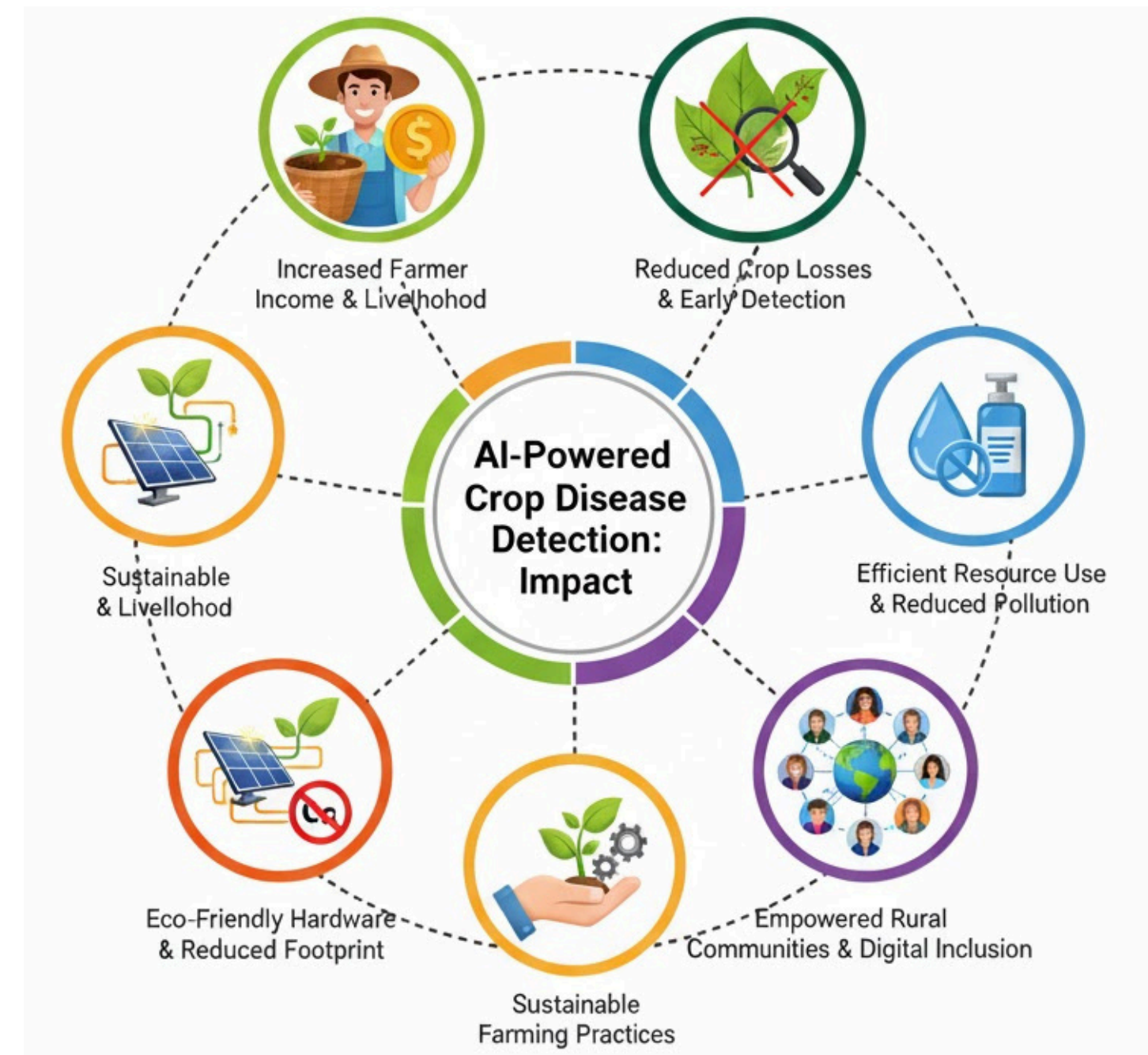
- Weather-integrated AI prevents over-irrigation, saving 20–25% water usage.
- Promotes crop rotation and green manure suggestions via AI dashboard.
- **Impact:** Supports climate-smart agriculture and long-term soil fertility.

Smart Transport & Pharmacy Connectivity

- Integrates local agri-pharmacies and transport networks to ensure timely medicine availability and doorstep delivery of eco-safe remedies, reducing treatment delays and supporting rural supply chain sustainability.

Why This Product is Essential:

1. Early Disease Detection = Higher Yields
2. Affordable Precision Agriculture for All



7. Financials



Estimated Prototype Development Cost

COMPONENTS	PRICE
Raspberry Pi 5 / Pi 4	4500
Camera Module (IMX708)	2500
Stepper Motor + Driver (28BYJ-48 + ULN2003)	350
Spectral Sensor (AS7341)	900
“VOC Sensor – SGP30 / MQ-135 (Air Quality Monitoring)”	250
Soil Sensor Kit (pH + Moisture + NPK)	600

SIM7600 GSM/4G Module	800
Temperature & Humidity Sensor (DHT22 / BME280)	200
Wires, PCB, connectors, LEDs	500
Software Open Source	FREE
TOTAL	~(10000-12000)

“All components selected are cost-efficient, readily available, and compatible for real-time AI-based disease detection.”

Experimentation Results



Experimental Setup

Location - Local test farm / agriculture field (e.g., Coimbatore, Tamil Nadu)

Crops Tested - Tomato, Paddy, and Cotton

Devices Used - 1 portable AI-powered prototype (Raspberry Pi + sensors)

Environment - Outdoor natural light, variable humidity (45–70%)

Data Collected - RGB images, spectral data, VOC readings, soil & weather data

Reference - Verified by agronomists for disease confirmation

Test Case	Detected Result	Ground Truth
Tomato Leaf Spot	Detected: Early Blight (94%)	Confirmed: Early Blight
Paddy Leaf	Detected: Healthy (98%)	Confirmed: Healthy
Cotton Leaf Curl	Detected: Leaf Curl Virus (91%)	Confirmed: Leaf Curl